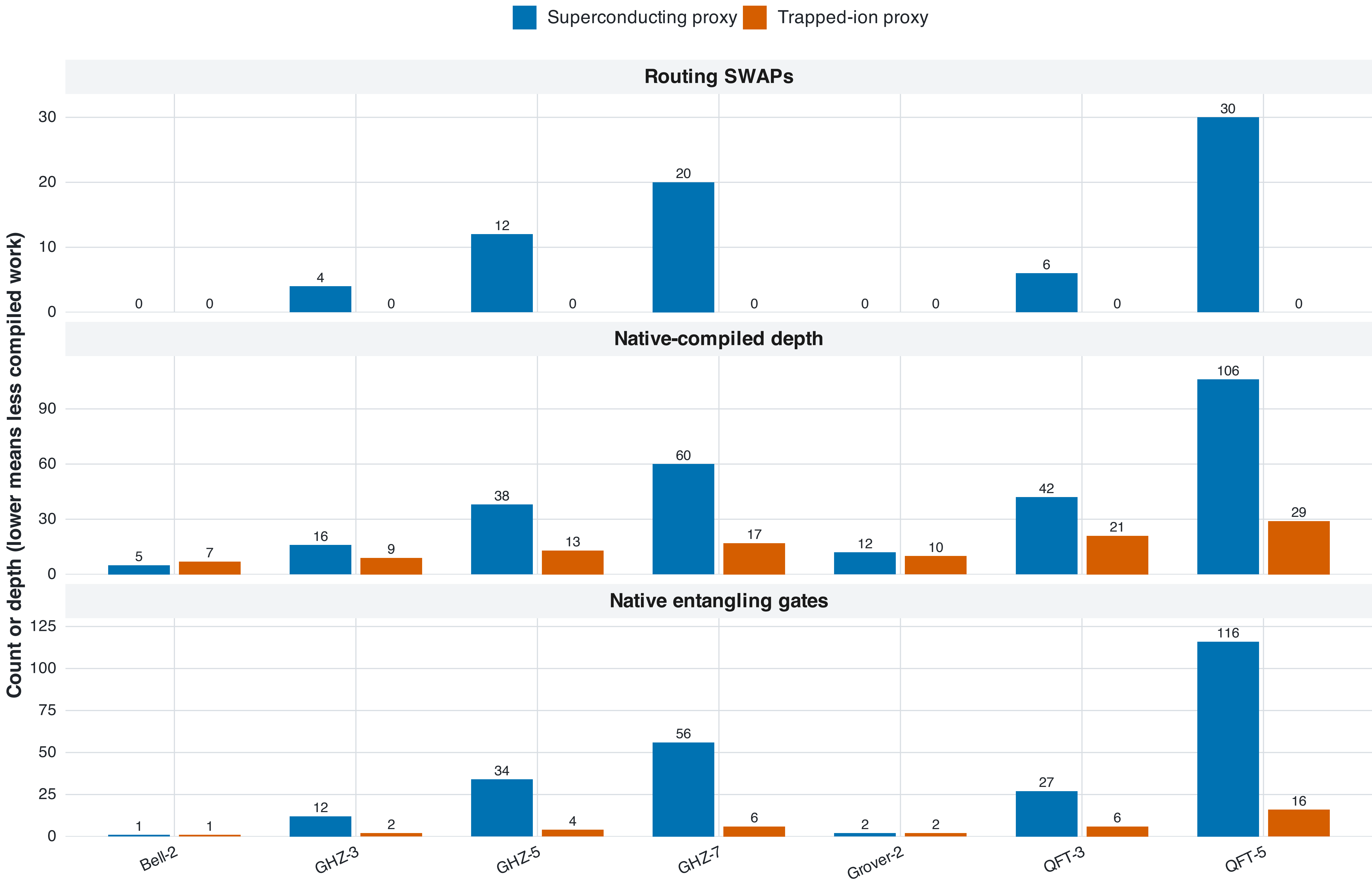
The Same Circuit Takes Different Roads

Controlled architecture proxies: identical logical circuits, different connectivity and native gates



Evidence class: offline architecture proxy. These values are not physical-device measurements.

- QUESTION — How do identical logical circuits change after routing and native-gate decomposition?
- READ — Compare the blue and orange bars within one circuit; shorter bars mean less compiled work.
- RESULT — GHZ and QFT expose the largest gap: line connectivity needs SWAP detours, while all-to-all connectivity needs none here.
- WHY IT MATTERS — Connectivity can change the practical circuit long before a quantum algorithm reaches a device.
- BOUNDARY — This answers the controlled proxy question; it does not rank physical IBM and Quantinuum QPUs.

Source: results/tables/architecture_validation_table.csv